

IRWI

The Irrigation Water Information Application



Research and Innovation for Water Security
Driving Action • Propelling Change



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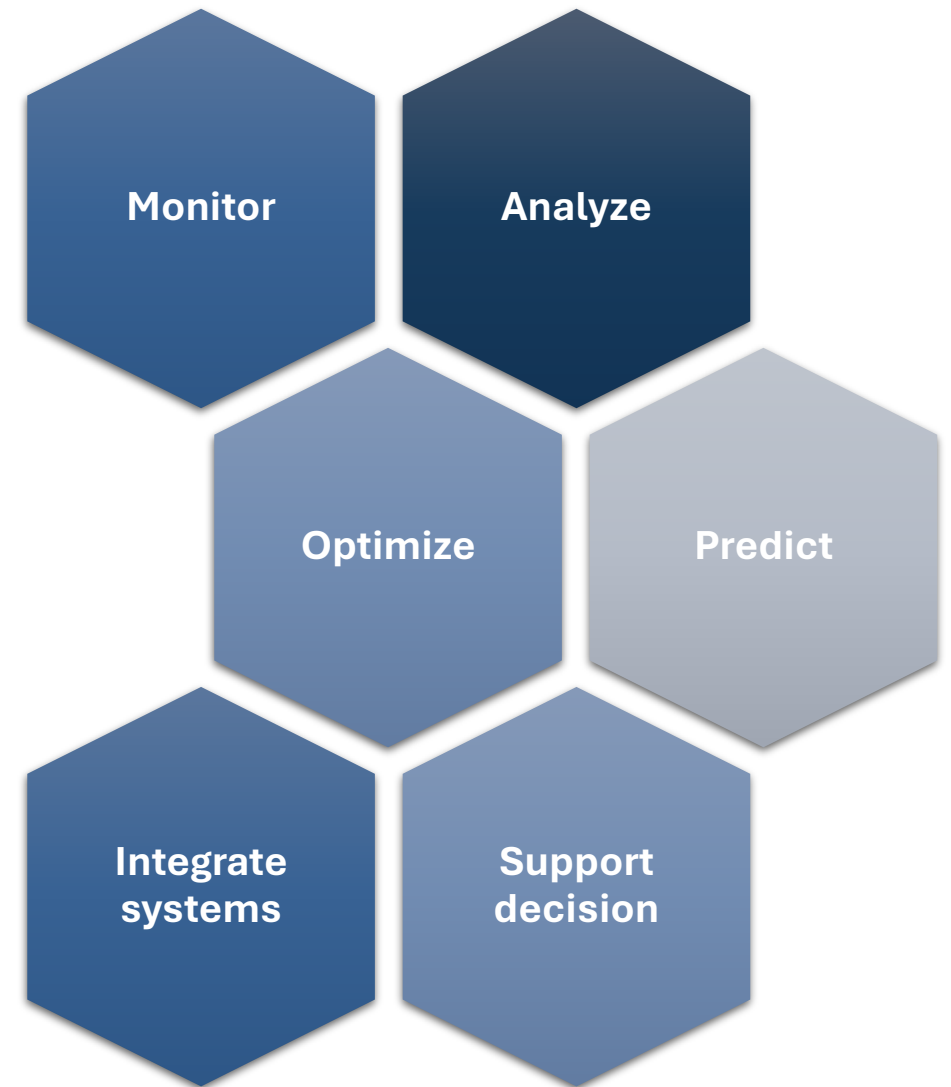
Global Challenge – Water, Land & Food Nexus

- Rising global demand
- Water scarcity intensifying
- Land degradation
- Climate change pressures
- Competing demands
- Nexus challenge

Key message: Business-as-usual resource use is no longer sustainable

Egypt Context & Role of Technology

- Water scarcity
- Limited arable land
- Growing demand
- Inefficiencies in water use:
- Need for optimization:



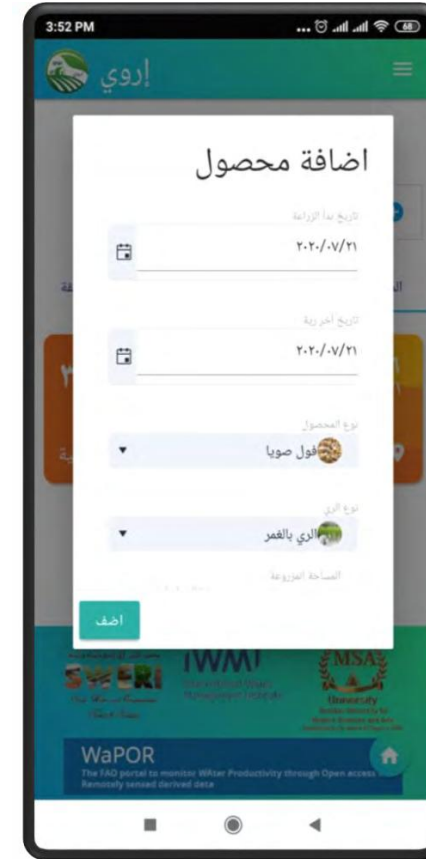
Key message: Business-as-usual resource use is no longer sustainable

IRWI

The Irrigation Water Information Application



irwicrop.com



- Soybean
- Maize
- Cotton
- Rice
- Potatoes

Objectives

Main Objective

- **Provide farmers with clear, timely guidance on irrigation**—when to irrigate, how much water to apply for each crop, and how long to operate irrigation systems.

Specific Objectives

- **Optimize irrigation scheduling:** Deliver precise crop water requirements and recommended irrigation intervals
- **Track crop performance:** Monitor crop growth stages and estimate productivity
- **Incorporate weather conditions:** Use real-time data on temperature and humidity to improve irrigation decisions
- **Enable informed management:** Provide easy access to irrigation history and daily performance records



Empowering farmers with simple, actionable information to improve water efficiency and crop productivity.

Testing, training and stakeholder engagement

- **Partnership with SWERI MSA University and Private sector:** Co-development and field validation with national experts
- **Field testing with farmers:** Direct visits, one-on-one training, and hands-on support
- **Capacity building workshops:** Trained farmers in Dakahlia & Kafr El-Sheikh on app use and installation
- **User-driven improvement:** Continuous feedback collected to refine features and usability
- **Farmer-informed design:** Requests integrated (e.g., pump power in HP, water source types)
- **Future services identified:** Pest management, disease prediction, and expanded extension support
- **Ongoing training (Phase II):** Capacity building on advanced tools (e.g., wheat model) for SWERI and private sector

IRWI app

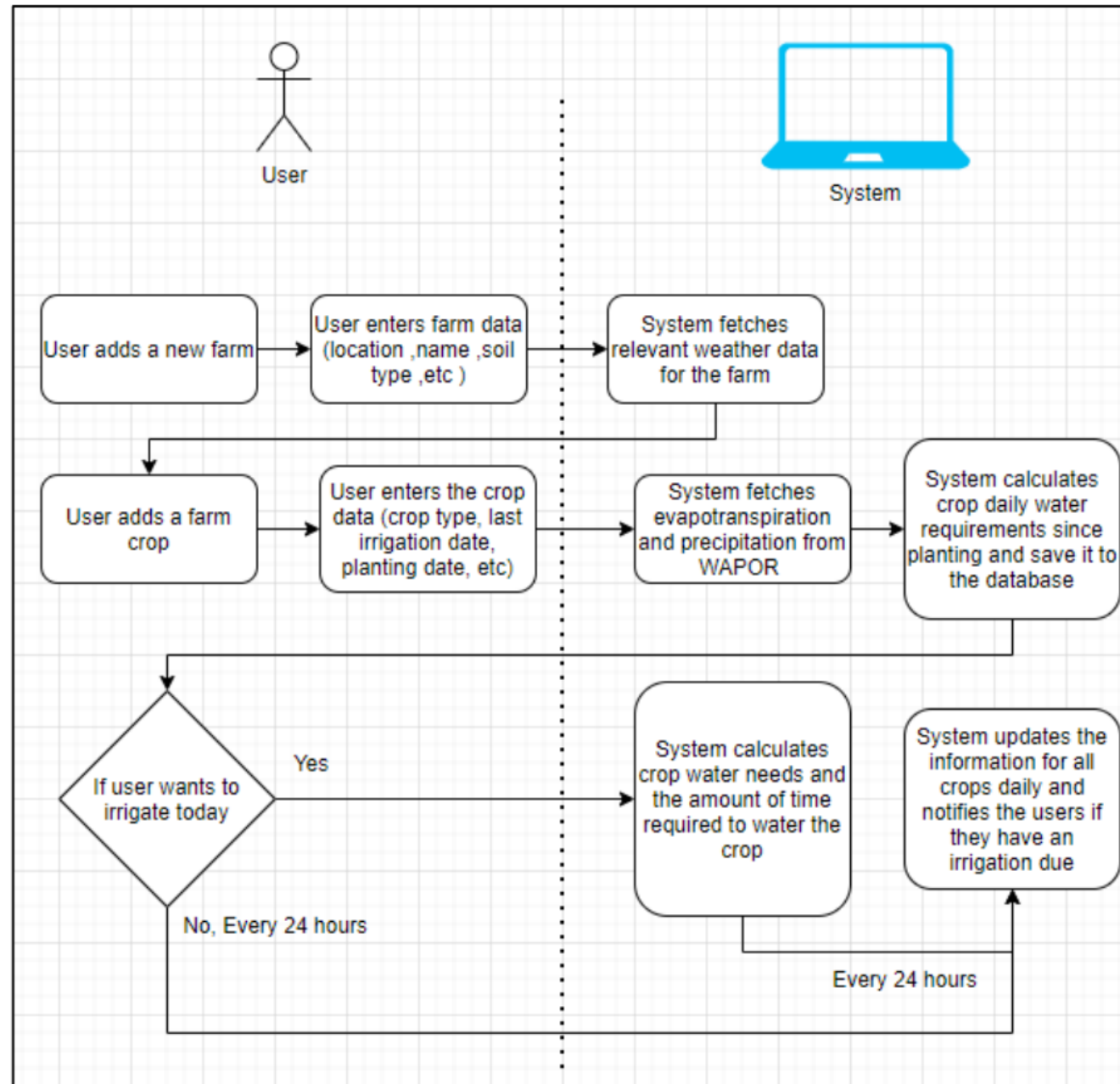
This website is designed using

- ASP.Net MVC-5
- bootstrap 4.0
- MS SQL Database

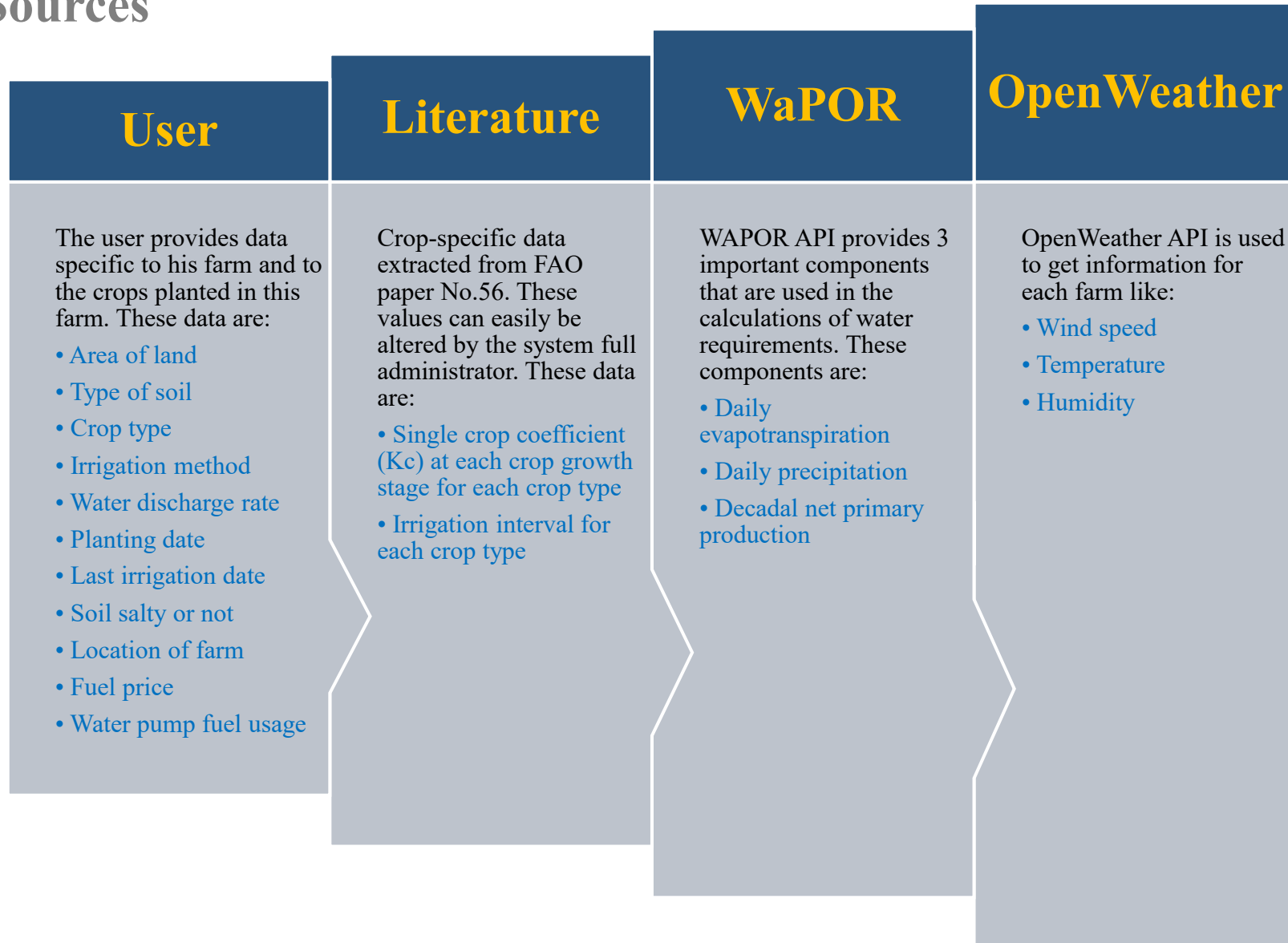
A mobile application is developed using Java



Workflow



Data Sources



Initial data used

Maize:

stage	Initial البداية	Mid النمو الخضري	Late التزهير	Ripe النضج	Stop irrigation
KC	0.6	1.2	1.5	0.6	20
No of days	20	30	40	30	
Irrigation interval	15				
Season days	Total number of days 120				

Rice:

stage	Initial البداية	Mid النمو الخضري	Late التزهير	Ripe النضج	Stop irrigation
KC	0.9	1.05	1.2	0.75	15
No of days	30	30	40	30	
Irrigation interval	6				
Season days	Total number of days 130				

Main system outputs

Weather information

٣١°

١٠ : ٥٤ : ٣٧
٢٠٢٠/٨/٧

الصغرى ٢٥

الإسكندرية 

Water requirements

اجمالي المياه المطلوبة للري
٧٥ متر مكعب

عدد ساعات تشغيل الطرمبه
٢ ساعة و ٦ دقائق

استهلاك الوقود
٣ لتر

سعر الوقود المستهلك
٥ جنيه

Ideal records

فول صويا تاريخ حصاد ٢٠٢٠/٨/٨		جدولة الري القياسية				
التاريخ	المرحلة	كمية المياه المستهلكة م مكعب	عدد ساعات التشغيل	العمر باليوم	استهلاك الوقود - لتر	سعر الوقود المستهلك
٢٠٢٠/٧/٢٦	النمو الخضري	٢٢٢,٠٣	٣٨,٥٥	١١	٦٦,٦٨	٩٨,٦٨

Practical irrigation

المراقبة		بيانات الموسم الحالي				الموسم السابق
ذرة تاريخ حصاد ٢٠٢٠/٨/٢٥		جدولة الري الفعلية				
التاريخ	المرحلة	كمية المياه المستهلكة م مكعب	عدد ساعات التشغيل	العمر باليوم	استهلاك الوقود - لتر	سعر الوقود المستهلك
٢٠٢٠/٠٨/٠٧	النمو الخضري	٧٤,٠١	٢,١١	٦	٣,٣٨	٥,٤٤

Harvested crops records

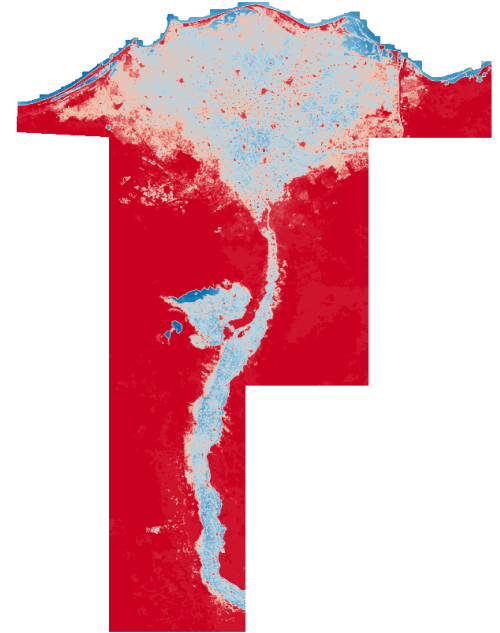
الموسم السابق		بيانات الموسم الحالي		المراقبة	
		بيانات يومية			
التاريخ	المرحلة	معدل البخر والنتج مم/يوم	معدل البخر والنتج للمحصول مم/يوم	معدل تساقط الامطار الفعلي مم	معامل المحصول

IRWI new developments

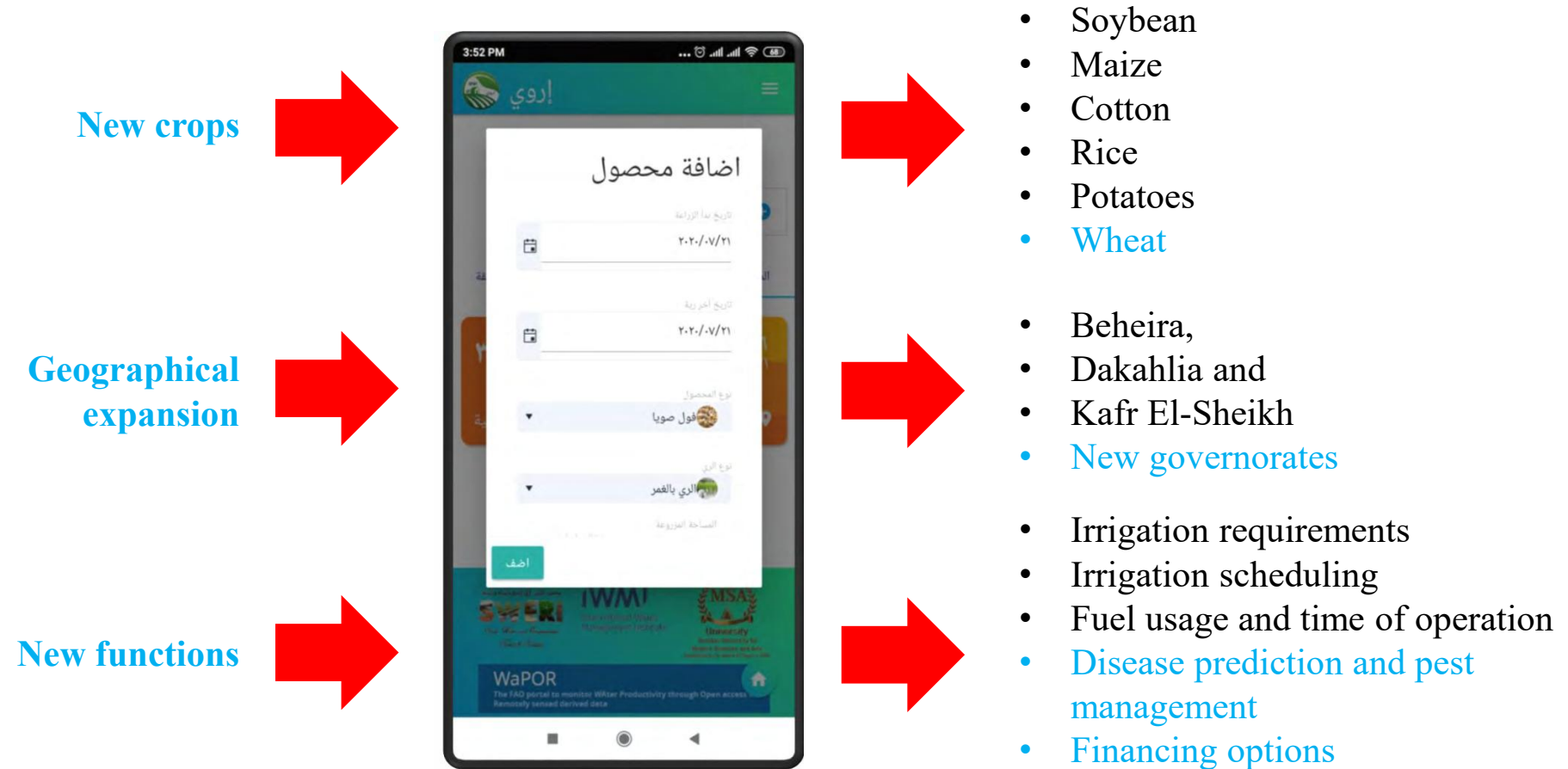
Irrigation scheduling model for wheat

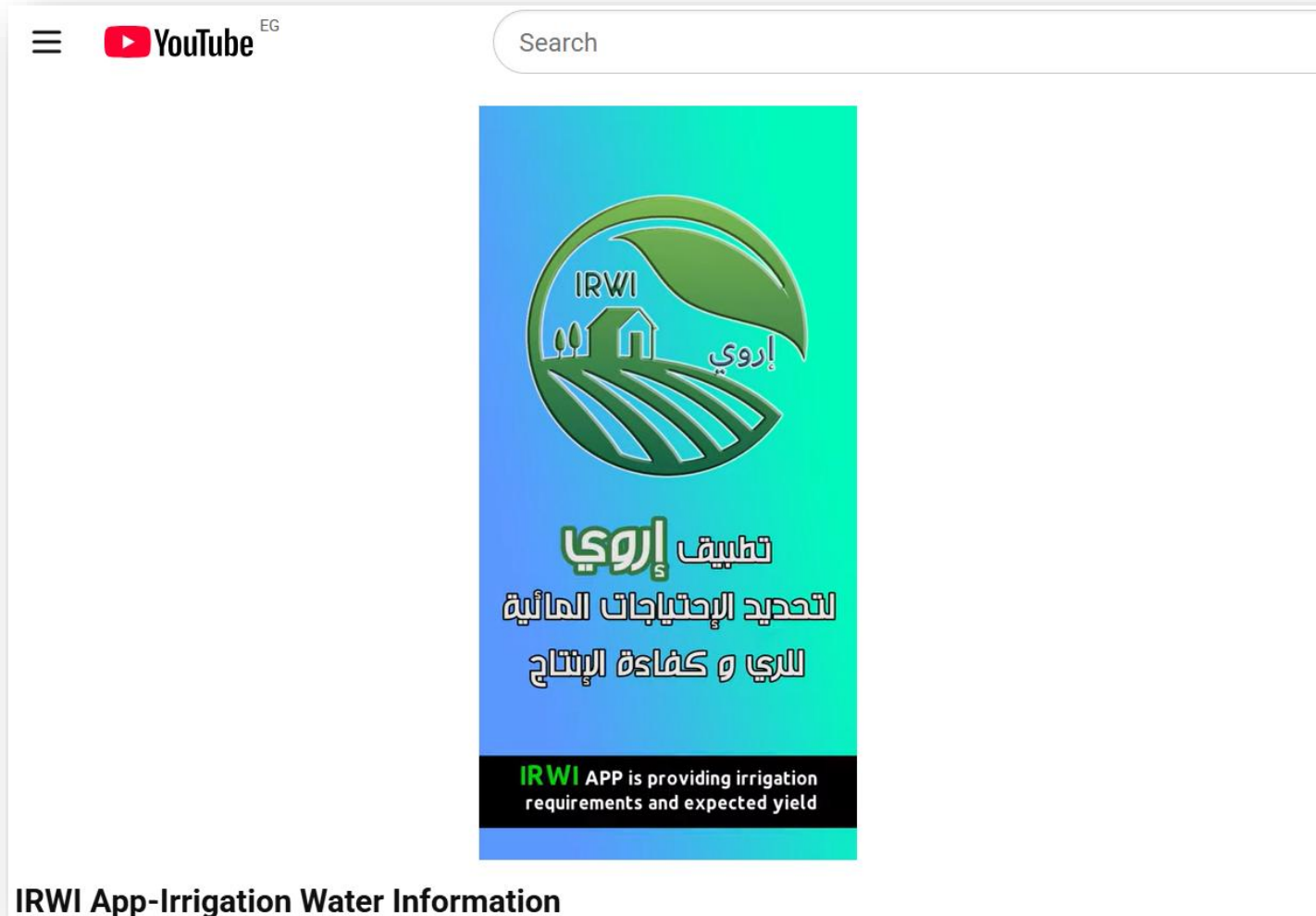
System workflow

- Geospatial setup and simulation parameters
- Input data specific to the intervention site
- Extraction of input data
- Disaggregation of dekadal WaPOR ETa to daily using ETref
- Assessment of crop coefficient (K_c) during WaPOR ETa gaps
- Water balance calculation
- Outputs visualization



Future development





<https://www.youtube.com/watch?v=teaCWyG0PAA>

www.iwmi.org

**THANK
YOU!**